VISUAL QUESTION ANSWERING FROM REMOTE SENSING IMAGES

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ABSTRACT

Remote sensing images carry wide amounts of information beyond land cover or land use. Images contain visual and structural information that can be queried to obtain high level information about specific image content or relational dependencies between the objects sensed. This paper explores the possibility to use questions formulated in natural language as a generic and accessible way to extract this type of information from remote sensing images, i.e. *visual question answering*. We introduce an automatic way to create a dataset using OpenStreetMap[1] data and present some preliminary results. Our proposed approach is based on deep learning, and is trained using our new dataset.

Index Terms— Visual Question Answering, Deep Learning, Natural Language, Remote Sensing, OpenStreetMap

1. INTRODUCTION

Visual Question Answering (VQA) is a relatively new task in the computer vision community, aiming at answering free-form and open-ended questions formulated in natural language about a given image. It has been introduced in its current form in [1]. As it deals with open-ended questions, it has the potential to provide a generic solution to many widely-studied tasks in the remote sensing community, as well as to answer very specific problems as illustrated in Figure 1.

While many studies have been dedicated to extracting information from remote sensing images, they are generally restricted to a specific problem (which means that extracting slightly different information will require in the case of supervised methods a drastically different approach trained with new labeled data) and are hardly usable without advanced image processing or coding skills. VQA holds a great promise: starting from a question formulated in natural language, a model can understand it and provide an answer (also in natural language). A working VQA system for remote sensing data would give an opportunity to enable information extraction from remotely-sensed images by entities not able to finance studies for a specific question as well as lifting the technical barriers that some interested users can face.

[1] www.openstreetmap.org



Problem	Example question
Classification	”Is it a rural area?”
Detection	”Is there a river?”
Counting	”How many buildings are there?”
Mix	”Is there a building next to the river?”

Fig. 1. Examples of questions corresponding to classical tasks and a mix of tasks

Thanks to the recent advances in deep learning, it is now possible to accurately predict answers to questions relating to images. An example of an application of VQA is to help visually impaired people [2]: through VQA they can ask questions about things that they see only partially and obtain clarifications helping them navigate or making choices. Similarly, remote sensing images could be more used by non-specialists if it was possible to query their content on demand and by questions formulated in natural language. The purpose of this article is to introduce a first database which can be used to train and evaluate VQA models focusing on remote sensing information and to present some preliminary results. To our knowledge, this is the first work adapting this idea to remote sensing data. This database has been created automatically by leveraging volunteered annotations from OpenStreetMap (OSM) and using Sentinel-2 images.

We present the procedure used to create the dataset in section 2. In section 3, we present a simple VQA model trained using our dataset and discuss first results achieved by our model.

2. THE DATASET

As discussed in the introduction, VQA is generally tackled using deep learning methods. Hence, the dataset is of prime importance, as it will be used both for the training and the evaluation of the methods. To develop our dataset focused on answering questions from remotely sensed data, we leveraged the geo-localized annotations from OSM. In this section we present how we constructed the dataset and analyse its structure and content.

2.1. Creation of the dataset

The dataset generation approach we followed is inspired by [3], which built a synthetic VQA dataset in an automated way. For a given image (in our case, acquired by Sentinel-2), the first step is to extract geo-localized OSM features that intersect with its extent. The questions are then generated solely based on these ground-truth features using the two main components detailed in the following paragraphs.

Component 1: construction of objects

To construct questions, we first need to define the objects of interest. An object of interest is the topic the question will be about: it is an urban element (e.g. "a road", "a building"), possibly filtered by some attributes (e.g. "a large building") or positional statements (e.g. "a road at the right of a field").

To construct an object, we first select a layer from OSM, among the following: *roads*, *water areas*, *buildings*, and *land use*. In the cases of *land use* and *buildings*, we use one of the possibilities in the "type"[2] field of the OSM feature. In summary, the base objects used to create the question objects can either be "road", "water area" or a specific type of building (e.g. "supermarket", "industrial building", ...) or of land use (e.g. "forest", "residential area").

The second step to create an object is to filter the base objects defined above. Three filters are used:

- **Area**: we filter objects based on their area. In the current version, we used hard thresholds to define "small" (less than $3000\mathrm{m}^2$), "medium" (less than $10000\mathrm{m}^2$) and "large" (more than $10000\mathrm{m}^2$) objects.
- **Shape**: we filter objects based on shape attributes (square, rectangular, circular), which are defined using basic geometrical properties (e.g. area-to-perimeter ratio).
- **Relative location**: we also implemented a way to filter objects based on their location relative to one another. 5 relations are defined ("at the top of", "on the right of", "at the bottom of", "on the left of" and "next to"). In the current database creation workflow, we select two different base objects. The first base object is filtered according to the relation. The centers of the bounding boxes enclosing the two objects are used to establish the relations. The specific case of "next to" is implemented as a hard threshold (less than 1000m away).

At the end of this filtering step, we obtain an object (e.g. "small industrial buildings next to a road") and the number of corresponding instances in the image (e.g., five).

Component 2: construction of questions

The questions are then constructed based on these objects and question templates. We define four types of questions of interest: presence, count, comparison, rural/urban classification.

In the first two cases, defining the answer is trivial from the number of instances of the object of interest obtained above. In the case of comparison, two different objects are created, and the number of instances is compared; in this case, three subquestions are used: "less", "more", "equal". Finally, we decided to answer the rural/urban classification based on a hard threshold on the number of buildings.

The generation of the questions is a stochastic process: the objects (both for the base objects and filters) are randomly selected, as well as the questions. However, we make sure that the rural/urban classification question is asked exactly once per tile.

2.2. Discussion

The proposed methodology allows to efficiently leverage open-access data to build a large dataset. From 9 Sentinel-2 images, we extracted 772 tiles of size 256×256 on which 77'232 question/answer couples are derived automatically. An advantage of the proposed method is that it allows us to easily control (via the distributions in the stochastic process) the distribution of the questions: we decided to have a similar amount of questions about objects' presence/absence and quantity (29.6% and 29.9% respectively), while the majority of questions dealt with comparisons (39.5%). Finally, 1% of the questions are about urban/rural classification of the area (a single question of this type is asked for each image). Controlling the distribution of question is less straightforward in the traditional approach (where humans freely asked questions about a given image). In this case, our approach limits the variety in the questions: the type of questions is fixed beforehand.

For similar reasons, we have a limited set of answers compared to traditional VQA datasets. Most of the questions (around 70%) can either be answered by "yes" or "no" (questions about the presence and the comparison of the number of objects), rural/urban classifications are answered by one of the two while the rest of the questions should be answered by

[2]The "type" field is used in the OSM layers to define the building usage or the type of land use of a given instance

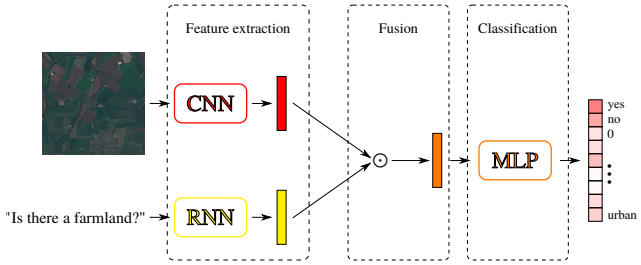


Fig. 2. Visual Question Answering model. In our experiments, the CNN is a ResNet-50 model [4] with a two layers fully-convolutional network to produce the feature vector, the RNN is a skip-thoughts model [5] and the fusion is a simple point-wise multiplication.

numbers. As this create a huge imbalance problem, we decided to cluster numerical answers to five ranges: 0; between 1 and 10; between 11 and 100; between 101 and 1000 and more than 1000, giving a set of 9 possible answers.

Finally, we decided to use Sentinel-2 images, making the dataset easily shareable. However, this means that the images have a resolution of 10m, making some features (such as roads or small buildings) present in OSM data hard to distinguish. Even though the dataset is based on Sentinel-2, the methodology presented in this article could be applied to any geo-localized remotely sensed images in future works.

3. EXPERIMENTS

We experimented using a simple model on our dataset. The model is described in subsection 3.1, while the results are in subsection 3.2.

3.1. Model

VQA models, in their simple form, extract visual features from the image as well as textual features from the question separately. These features are then merged to perform a classification between the possible answers [6]. The visual feature extraction is generally based on a Convolutional Neural Network (CNN) and the textual feature extraction on a Recurrent Neural Network (RNN). This architecture is shown in Figure 2.

In our experiments, the CNN is based on a ResNet-50 model [4] which has been pre-trained on ImageNet [7] followed by two fully-connected layers (from size 8192 to 2048 and from 2048 to 1200). The RNN is the pre-trained skip-houghts model [5] followed by a fully-connected layer (from size 2400 to 1200). The fusion is a simple point-wise multiplication between the two feature vectors and the classification is done by two fully-connected layers (from size 1200 to 256 and 256 to the number of class, i.e. 9 using the proposed

Type	Accuracy
Presence	87.02%
Count	69.68%
Comparison	83.78%
Rural / Urban	80.52%
Average	80.24%
Overall	80.55%

Table 1. Accuracy of the proposed method

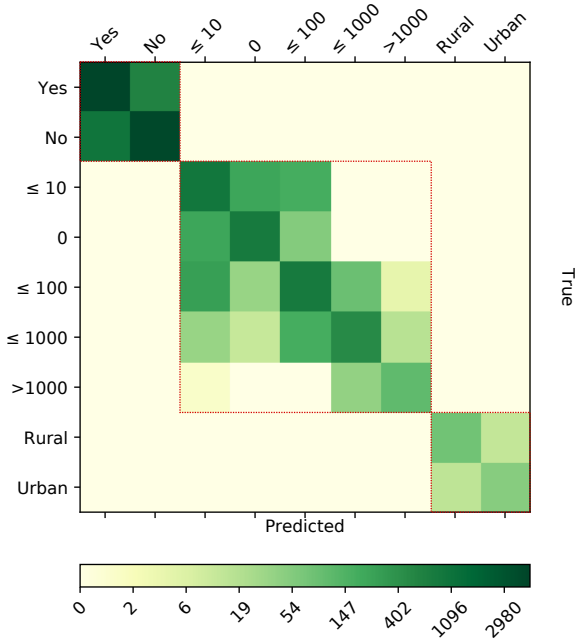


Fig. 3. Confusion matrix of the answers provided by the proposed model on the test set (logarithm scale). Dashed lines group answers by type ("Yes/No", numbers and "Rural/Urban").

dataset). We used a dropout of 0.5 for the fully-connected layer during the training of the model.

We train the model using the Adam optimizer [8] with a learning rate of 10^{-5} for 150 epochs.

3.2. Results and discussion

We report the accuracy of our model (given by the ratio of correct answers) for the different categories of questions in Table 1 as well as the confusion matrix in Figure 3. We also show some visual results in Figure 4. The proposed model achieves about 80% accuracy (both overall and average), indicating that visually answering questions based on remote sensing data is possible, even though there is definitely room for improvement. The visual examples of Figure 4 show that our VQA model can answer non-trivial questions, but tends to struggle on questions dedicated to small elements, possibly



Question	What is the number of farmlands?	Is there a small forest?	Is a medium building present?
Prediction	Between 10 and 100	Yes	No
Ground Truth	Between 10 and 100	Yes	Yes

Fig. 4. Sample questions from the test set.

due to the resolution of Sentinel-2 images. When looking at the accuracy for the different types of questions (in Table 1), it is quite clear that the most difficult task is the one of counting, even though we simplified it (see subsection 2.2). This is not surprising as the task of counting is notoriously hard for VQA models [9]. The confusion matrix presented in Figure 3 shows that our model tends to provide logical answers even when committing mistakes (e.g. not answering "no" to "How many ..."). Despite this being encouraging, it might also be explained by the rather limited variety of questions types present in the dataset.

4. CONCLUSION

We created the first dataset for Visual Question Answering from remote sensing images as a first step towards a generic and easy information extraction tool from satellite data. Preliminary results show encouraging results (80% accuracy) suggesting that there is promise for the application of such systems in the future. Future work should be dedicated to improving the dataset to include more variety in the questions, with potentially high-resolution data. The presented method for dataset creation can be adapted to achieve this objective. There is room for improvement concerning the model, for instance using an attention mechanism [6], which could improve both the performances and the interpretability of the results, or introducing modular components to improve performances on tasks where our current model performs the worst (e.g. for counting, a CNN as proposed in [10] might be used instead).

5. REFERENCES

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